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In [1]: #THUSHAR K P, ROLL : B24CTA60
import numpy as np
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt
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In [2]: data=pd.read_csv("weather.csv")
display(data)
```

	Day	Outlook	Temperature	Humidity	Wind	Play Tennis
0	D1	Sunny	Hot	High	Weak	No
1	D2	Sunny	Hot	High	Strong	No
2	D3	Overcast	Hot	High	Weak	Yes
3	D4	Rain	Mild	High	Weak	Yes
4	D5	Rain	Cool	Normal	Weak	Yes
5	D6	Rain	Cool	Normal	Strong	No
6	D7	Overcast	Cool	Normal	Strong	Yes
7	D8	Sunny	Mild	High	Weak	No
8	D9	Sunny	Cool	Normal	Weak	Yes
9	D10	Rain	Mild	Normal	Weak	Yes
10	D11	Sunny	Mild	Normal	Strong	Yes
11	D12	Overcast	Mild	High	Strong	Yes
12	D13	Overcast	Hot	Normal	Weak	Yes
13	D14	Rain	Mild	High	Strong	No

```
In [3]: le=LabelEncoder()
for column in data.columns:
    data[column]=le.fit_transform(data[column])
display(data)
```

	Day	Outlook	Temperature	Humidity	Wind	Play Tennis
0	0	2	1	0	1	0
1	6	2	1	0	0	0
2	7	0	1	0	1	1
3	8	1	2	0	1	1
4	9	1	0	1	1	1
5	10	1	0	1	0	0
6	11	0	0	1	0	1
7	12	2	2	0	1	0
8	13	2	0	1	1	1
9	1	1	2	1	1	1
10	2	2	2	1	0	1
11	3	0	2	0	0	1
12	4	0	1	1	1	1
13	5	1	2	0	0	0

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In [4]: x=data.drop(['Play Tennis'],axis=1)
y=data['Play Tennis']
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In [5]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=1,random_state=42)
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In [6]: tree=DecisionTreeClassifier(criterion="entropy")
tree.fit(x_train,y_train)
```

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Out[6]: DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy')
```

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In [7]: y_pred=tree.predict(x_test)
display(y_pred)
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array([1])
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In [8]: acc=accuracy_score(y_pred,y_test)
display(acc)
```

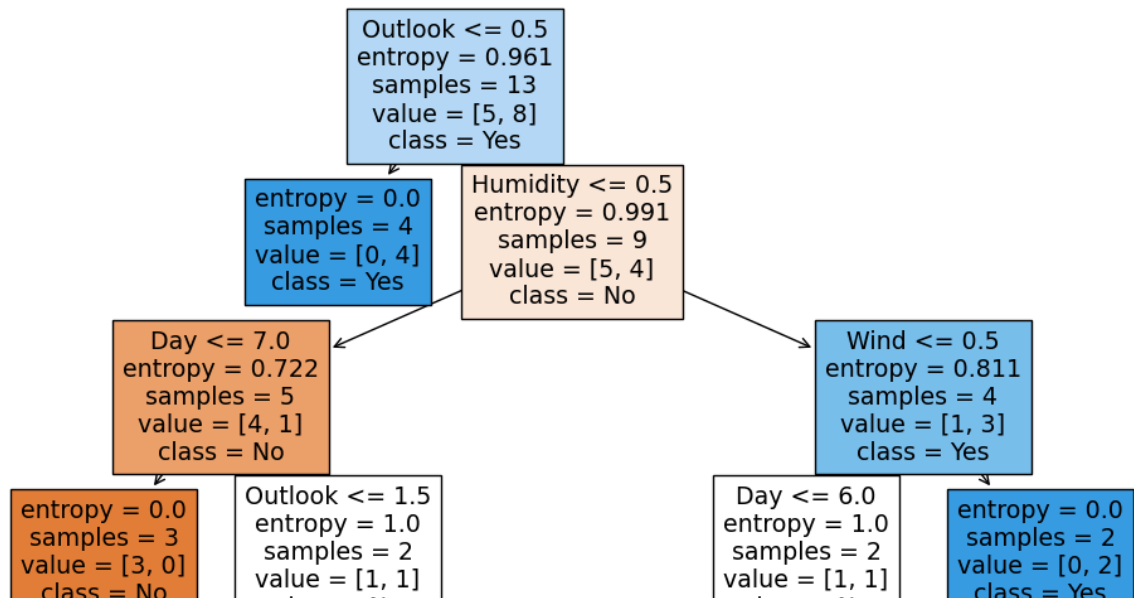
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1.0
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In [9]: splittingfeat=tree.tree_.feature
display(splittingfeat)
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array([ 1, -2,  3,  0, -2,  1, -2, -2,  4,  0, -2, -2, -2], dtype=int64)
```

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In [10]: plt.figure(figsize=(12,8))
plot_tree(tree,feature_names=list(x.columns),class_names=['No','Yes'], filled=True)
plt.show
```

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Out[10]: <function matplotlib.pyplot.show(close=None, block=None)>
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